Mathematical Analysis 2

Seminar 9

Integrals with parameters

[from high school]

If $g(t) = \int_a^t f(x)\,\mathrm{d}x$, then $g'(t) = f(t)$. [f is continuous]

[the general case]

If $g(t) = \int_{a(t)}^{b(t)} f(t,x)\,\mathrm{d}x$, then

$$g'(t) = b'(t) \cdot f(t, b(t)) - a'(t) \cdot f(t, a(t)) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(t,x)\,dx$$

a, b differentiable functions

$\frac{\partial f}{\partial t}$ exists and is continuous.

Ex. 1 If $g(t)=\int_{t^2}^{t^3} f(2x+t^2)\,dx$, then find $g'(t)$

$$g'(t)=\underbrace{3t^2}_{b'}\cdot f(\underbrace{2t^3}_{b}+t^2)-\underbrace{2t}_{a'}\cdot f(\underbrace{2t^2}_{a}+t^2)+\int_{t^2}^{t^3}\underbrace{f'(2x+t^2)\cdot 2t}_{\frac{\partial}{\partial t}f(2x+t^2)}\,dx=$$

$$=3t^2 f(2t^3+t^2)-2t f(3t^2)+2t\underbrace{\int_{t^2}^{t^3} f'(2x+t^2)\,dx}_{I}$$

$$g'(t)=3t^2 f(2t^3+t^2)-2t f(3t^2)+$$
$$+t f(t^2+2t^3)-t f(3t^2)=(3t^2+t) f(t^2+2t^3)-3t f(3t^2)$$

$$I=\int_{t^2}^{t^3} f'(\underbrace{2x+t^2}_{y})\,dx$$

$y=2x+t^2$; $dy=2dx$

$$I=\frac{1}{2}\int_{3t^2}^{t^2+2t^3} f'(y)\,dy=$$

$$=\frac{1}{2}f(y)\Big|_{3t^2}^{t^2+2t^3}=\frac{1}{2}(f(t^2+2t^3)-f(3t^2$$

2nd method: $y=2x+t^2$; $dy=2dx \Rightarrow g(t)=\int_{3t^2}^{t^2+2t^3}\frac{1}{2}f(y)\,dy$

$$g'(t)=\frac{1}{2}\left[(t^2+2t^3)'\cdot f(t^2+2t^3)-(3t^2)'\cdot f(3t^2)\right]$$

$$=\frac{1}{2}\left((2t+6t^2)\cdot f(t^2+2t^3)-6t\cdot f(3t^2)\right)=\ldots \text{ (the same result)}$$

$\boxed{\textbf{Ex. 2}}$ If $g(t) = \int_0^t f(x)\cdot(t-x)^{n-1}\,\mathrm{d}x$ $(n \in \mathbb{N}^*)$, then find $g^{(n)}(t)$.

$$g'(t) = \underbrace{1\cdot f(t)\cdot(t-t)^{n-1}}_{0} - 0 + \int_0^t f(x)\cdot(n-1)(t-x)^{n-2}\,dx =$$

$$= (n-1)\int_0^t f(x)(t-x)^{n-2}\,dx$$

$$g''(t) = (n-1)\cdot(n-2)\int_0^t f(x)(t-x)^{n-3}\,dx \Rightarrow \dots \Rightarrow g^{(n-1)}(t) = (n-1)!\int_0^t f(x)(t-x)^0\,dx = (n-1)!\int_0^t f(x)\,dx$$

$$g^{(n)}(t) = (n-1)!\,f(t)$$

Ex. 3 Compute $\underbrace{\int_0^{\frac{\pi}{2}} \ln\left(\cos^2 x + a^2 \cdot \sin^2 x\right) \mathrm{d}x}_{I(a)\,=\,?}$ $\quad (a \geq 0)$.

$$
\begin{tikzcd}
\otimes \arrow[r, "\int dx"] \arrow[d, "\frac{d}{da}"'] & ? \\
{} \arrow[r, "\int dx"] & {} \arrow[u, "\int da"']
\end{tikzcd}
$$

$$I'(a) = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x \cdot 2a}{\cos^2 x + a^2 \cdot \sin^2 x}\, dx = 2a \int_0^{\pi/2} \frac{\tan^2 x}{1 + a^2 \cdot \tan^2 x}\, dx$$

$$\tan x = y \;;\; x = \operatorname{arctg} y \;;\; dx = \frac{1}{1+y^2}\, dy$$

$a \neq 1$

$$I'(a) = 2a \int_0^{\infty} \frac{y^2}{(1+a^2y^2)(1+y^2)}\, dy = 2a \cdot \int_0^{\infty} \frac{(1+a^2y^2) - (1+y^2)}{(1+a^2y^2)(1+y^2)} \cdot \frac{1}{a^2-1}\, dy =$$

$$= \frac{2a}{a^2-1} \int_0^{\infty} \left(\frac{1}{1+y^2} - \frac{1}{1+a^2y^2} \right) dy = \frac{2a}{a^2-1} \left(\arctan y - \frac{1}{a} \cdot \arctan(ay) \right) \Big|_0^{\infty} =$$

$a \neq 0$

$$= \frac{2a}{a^2-1} \left(\frac{\pi}{2} - \frac{1}{a} \cdot \frac{\pi}{2} \right) = \frac{\pi}{\cancel{2}} \cdot \frac{\cancel{2a}}{\cancel{(a-1)}(a+1)} \cdot \frac{\cancel{a-1}}{\cancel{a}} = \frac{\pi}{a+1} \longleftarrow \text{is defined also for } a \in \{0,1\}.$$

$$I'(a) = \frac{\pi}{a+1} \;\Big|\; \int da \;\Rightarrow\; I(a) = \pi \ln(a+1) + C$$

$$a := 1 \Rightarrow I(1) \begin{matrix} = 0 \\ = \pi \ln 2 + C \end{matrix} \Big\} \Rightarrow C = -\pi \ln 2$$

$$\boxed{I(a) = \pi \ln \frac{a+1}{2}, \quad a \geq 0}$$

$\boxed{\textbf{Ex. 4}}$ Compute $\underbrace{\int_0^{\frac{\pi}{2}} \frac{\arctan(a\cdot\sin x)}{\sin x}\,dx}_{I(a)}$ $(a \geq 0)$. (or, $a \in \mathbb{R}$)

$$I'(a) = \int_0^{\pi/2} \frac{1}{\sin x} \cdot \frac{1}{1+a^2\cdot\sin^2 x} \cdot \sin x \, dx = \int_0^{\pi/2} \frac{1}{1+a^2\sin^2 x}\,dx$$

$\tan x = y$; $x = \arctan y$; $dx = \frac{1}{1+y^2}\,dy$

$$y^2 = \frac{\sin^2 x}{\cos^2 x} = \frac{\sin^2 x}{1-\sin^2 x} \iff y^2 - y^2\sin^2 x = \sin^2 x \iff \sin^2 x = \frac{y^2}{1+y^2}.$$

$$I'(a) = \int_0^\infty \frac{1}{1+a^2\cdot\frac{y^2}{1+y^2}} \cdot \frac{1}{1+y^2}\,dy = \int_0^\infty \frac{1}{(1+y^2)+a^2y^2}\,dy = \int_0^\infty \frac{1}{1+(1+a^2)y^2}\,dy =$$

$$= \frac{1}{\sqrt{1+a^2}} \arctan \sqrt{1+a^2}\,y \Big|_0^\infty = \frac{\pi}{2}\cdot\frac{1}{\sqrt{1+a^2}}, \ \forall a \in \mathbb{R} \ \Big| \int da$$

$$\Rightarrow I(a) = \frac{\pi}{2}\cdot\ln\left(a+\sqrt{1+a^2}\right) + C$$

$$a=0 \Rightarrow \left.\begin{array}{l} I(0) = 0 \\ = \frac{\pi}{2}\ln 1 + C = C \end{array}\right\} \Rightarrow C = 0$$

$$\boxed{I(a) = \frac{\pi}{2}\ln\left(a+\sqrt{1+a^2}\right)}$$

Remark: $\displaystyle\int_0^{\pi/2} \frac{\arctan(\sin x)}{\sin x}\,dx = \frac{\pi}{2}\ln\left(1+\sqrt{2}\right)$